

EXCHANGE RATES AND HOW THEY ARE DETERMINED

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Exchange rate, which is expressed as Rupees per dollar or Rupees per Euro, is a price: price of Indian currency in terms of dollars or some other currency (just like the price of onion expressed in Rupees per kg of onion)

Exchange rates could be determined entirely by market forces. That is, the value of Rupee (or any other currency) would fluctuate depending on the demand for and supply of Rupee in the foreign exchange market (just as the demand for any other good, say onion, would vary depending on the demand for and supply of onion). We call such a system, a completely **flexible or floating system of exchange rate determination** ('clean floating' means fully determined by the market).

Exchange rates could be fixed at a certain level – a level that is determined by the government or the Central Bank. This is the **fixed exchange rate system**. The government or the Central Bank intervenes in the foreign exchange market to 'fix' the exchange rate at a certain level.

Most countries followed a fixed exchange rate system in the decades before World War I, between the mid-1920s and 1931, and between 1945 and 1973. Today, while most countries are committed to flexible exchange rates, Central banks of these countries intervene frequently in the exchange rate market to ensure that exchange rates do not fluctuate too much (resulting in 'managed floating' or 'dirty floating' of their currencies).

How does the foreign exchange market operate? Every time Indian goods are exported (or are purchased by foreigners), some trader somewhere in the world will have exchanged her dollars (or Euro or Yen) into Indian Rupees (so as to purchase the Indian goods). Similarly, when a foreign investor decides to set up a factory in India, she or he would again be exchanging her dollars with Rupees (so as to pay for the land to set up the factory in India, to pay Indian workers, and so on).

Therefore, exports of Indian goods or services or inflow of capital into the country in the form of investment or loans will have an impact on the foreign exchange market: a rise in the demand for Indian Rupees.

On the other hand, imports of goods or services into India or the outflow of capital from the country in the form of investment or loans will result in a fall in the demand for Indian Rupees.

In the foreign exchange market, if there is a rise in the demand for Indian Rupees (relative to say dollar) without any change in the supply of Indian Rupees, the result will be a rise in the value of Indian Rupees (vis-à-vis dollars). In other words, Rupee appreciates – from say Rupees 50 a dollar to Rupees 40 a dollar.

Why and how the Central Bank intervenes?

If the currency appreciates, it will have different impacts on different sectors of the economy. On the one hand, imports and foreign travel will cost less. Exchanging Rs.1 lakh, you will receive 2500 dollars if the exchange rate is Rs.40 per dollars – much better than the 2000 dollars you would have received if the exchange rate were Rs.50 per dollar.

However, exporters suffer due to appreciation of the domestic currency. A garment manufactured in Ludhiana and priced at Rs.1000 per piece in Indian currency would cost 25 dollars in the US market if the exchange rate appreciated to Rs.40 a dollar – had the exchange rate been Rs.50 a dollar, the cost would have been only 20 dollars. That would make Indian garments far less attractive in the export markets.

So if the exchange rate is completely flexible or determined entirely by market forces, one can see that the domestic currency will appreciate with rising export earnings. However, with currency appreciation, the country's exports will slow down (as it may lose its competitiveness in the export markets), and thereby foreign exchange earnings will come down.

Thus to summarize:

Growth in exports > Rise in foreign currency inflows > Appreciation of domestic currency > slow down in the growth of exports > Fall in foreign currency inflows

However, many developing countries (China) may not want appreciation in domestic currency to put a break to its export demand growth. This they achieve by intervening in the foreign exchange market in an appropriate way.

Even after shifting to an exchange rate system that is flexible (and hence determined more by market forces), the Central Banks in India, China and many other countries intervene in the foreign exchange market. This is to ensure that the currency does not appreciate or depreciate to a level that might hurt the economy.

How can the Central Bank intervene to avoid currency appreciation during a period of rising inflows of foreign exchange?

When the demand for Indian currency rises (following a rise in export demand), the Central Bank intervenes to avoid appreciation. Its intervention will be in the form of increasing the supply of Indian Rupees – so that the value of the Indian Rupee does not go up due to the rise in demand.

The Central Bank buys dollars and issues Rupees against such dollars.

That is, when market players try to:

Sell dollars and buy Rupees, the Central Bank intervention will be to

Buy dollars and sell Rupees.

In this way, the Central Bank ensures adequate supply of Rupees to meet the increased demand for Rupees from market players. The Central Bank intervention thus averts an appreciation of the Rupee.

Central Bank Balance Sheet

Assets	Liabilities
Foreign Assets ---	Currency in circulation --
Domestic Assets --	Deposits held by private banks with the Central Bank --

Foreign assets consist mainly of foreign currency bonds owned by the Central Bank. They also make up the Central Bank's official international reserves.

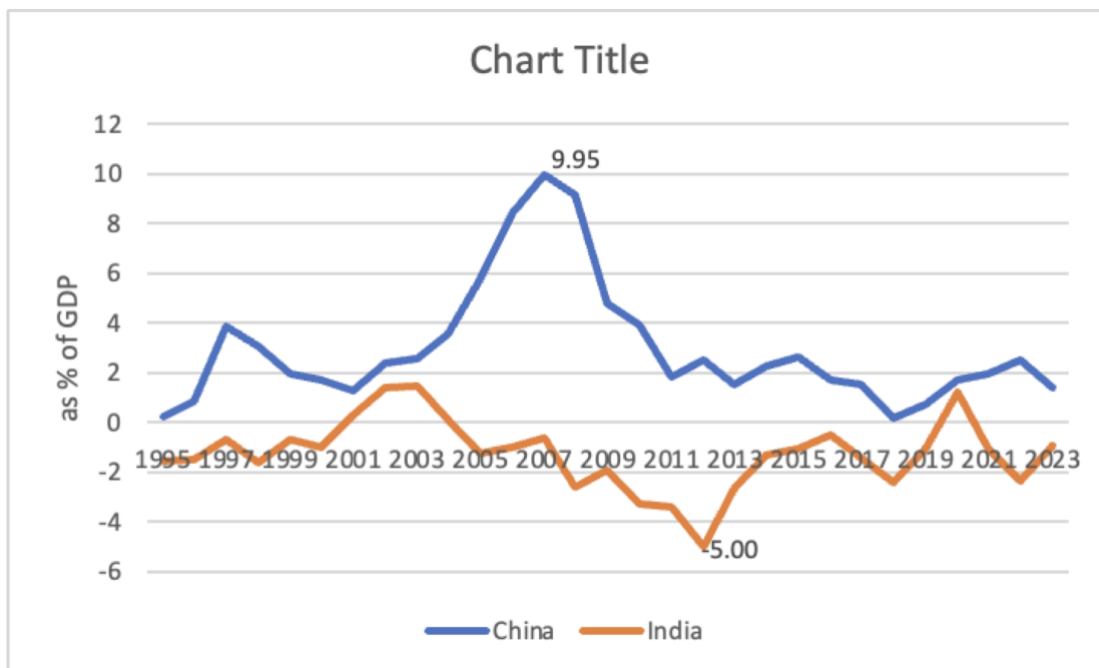
Domestic assets consist of Central Bank's credit to commercial banks, Central Bank ownership of government bonds (in other words credit to the government), and Central Bank advances to the commercial sector.

When the Central Bank acquires foreign assets (say US treasury bonds worth \$100), then the Central bank assets go up by \$100. Its liabilities too rise by an equivalent amount (rise in currency in circulation if the payment for foreign assets is made in cash; rise in deposits by private banks if the payment is made through cheque).

Note that currency in circulation and Deposits held by private banks with the Central Bank are part of the reserve money. (Remember: money supply/reserve money = money multiplier).

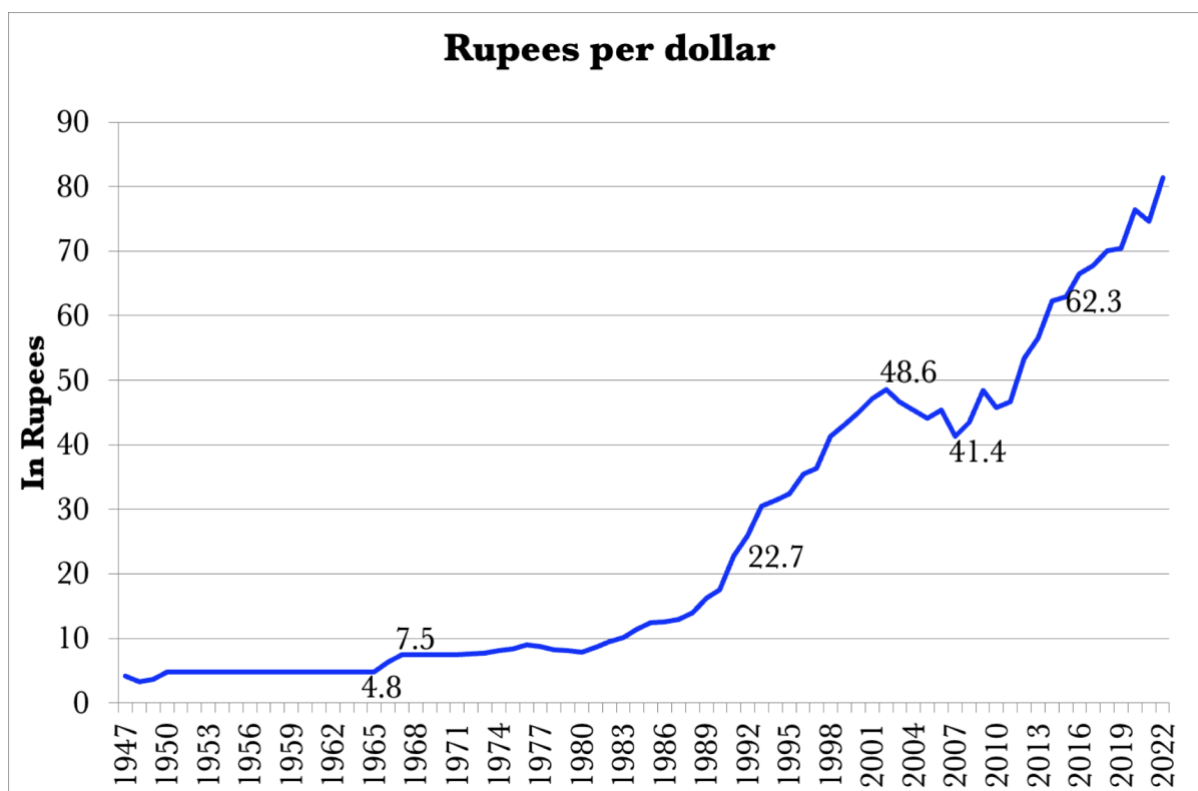
Therefore, purchase of foreign assets or domestic assets by the Central Bank will cause an increase in domestic money supply.

Alternatively, sale of foreign assets or domestic assets by the Central Bank will cause a decrease in domestic money supply.



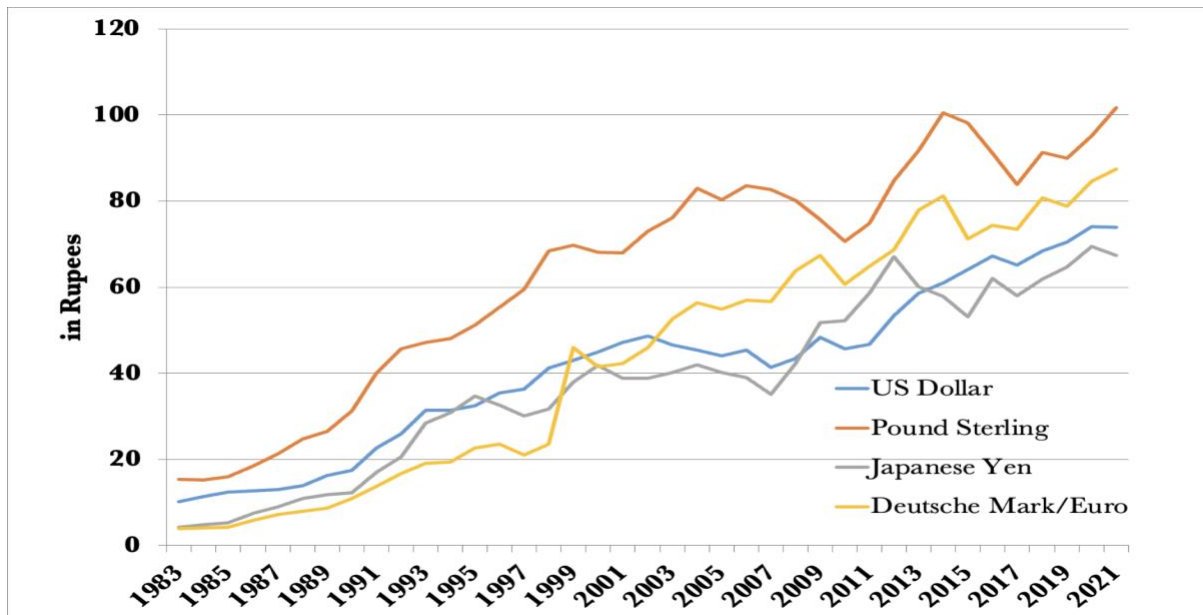
Current account balance as % of GDP: China and India, 1995-2023

Source: World Development Indicators



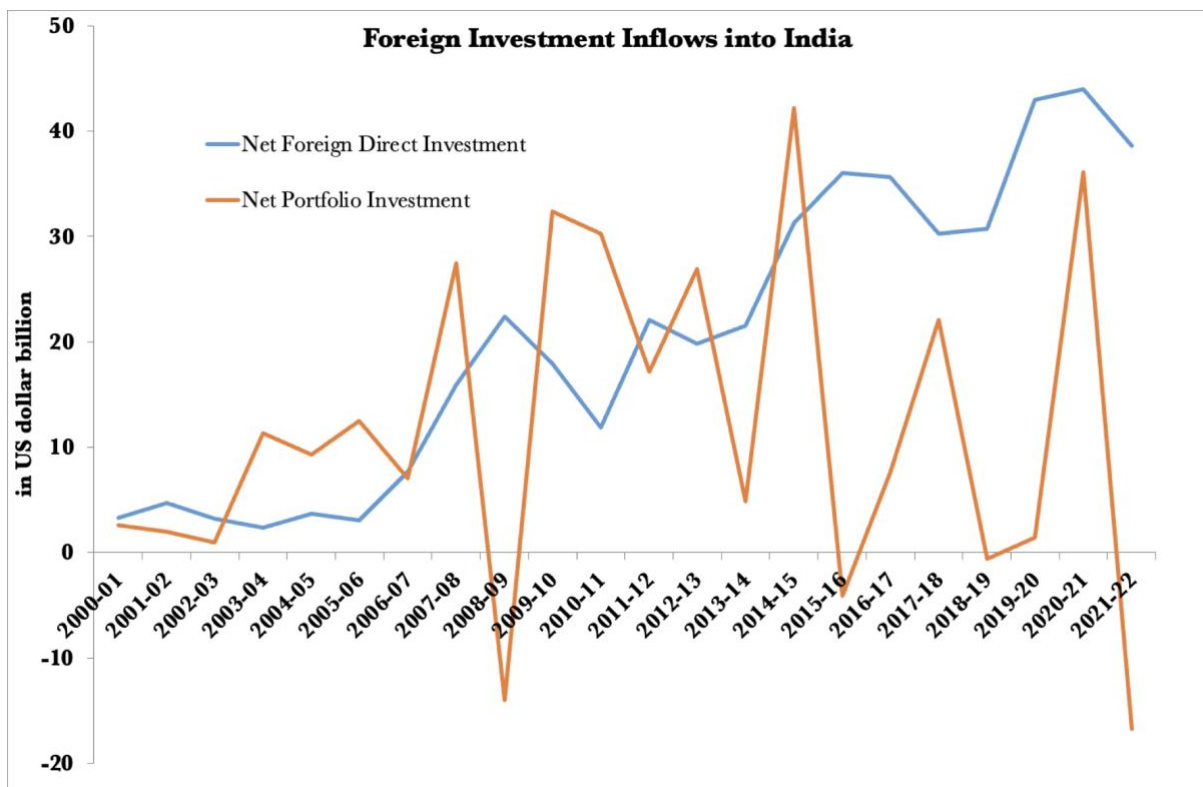
Exchange Rate: Indian Rupees per US dollar, 1947-2022

Source: <https://www.bookmyforex.com/blog/1-usd-inr-1947-till-now/>



Exchange Rate: Indian Rupees per unit of foreign currency

Source: Reserve Bank of India, Handbook of Statistics on the Indian Economy, Table 136



Net Foreign Direct Investment (FDI) and Net Foreign Portfolio Investment (FPI) into India

Source: Reserve Bank of India, Handbook of Statistics on the Indian Economy, Table 145



Inflows of Foreign Portfolio Investment (FPI) into India and Rupee-dollar Exchange Rate, 2005-2012

Source: Reserve Bank of India, Handbook of Statistics on the Indian Economy, Table 145

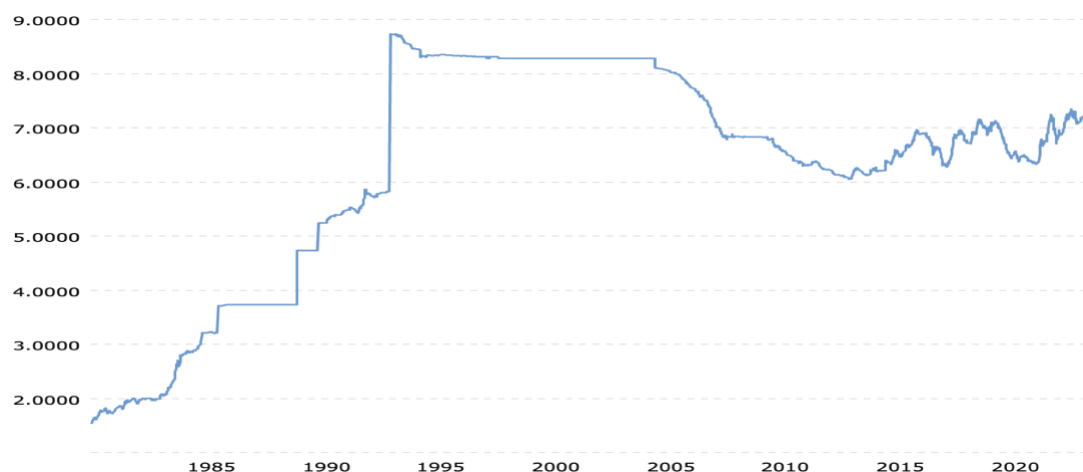
China's holdings of US Treasury securities

(in billions of US dollars)



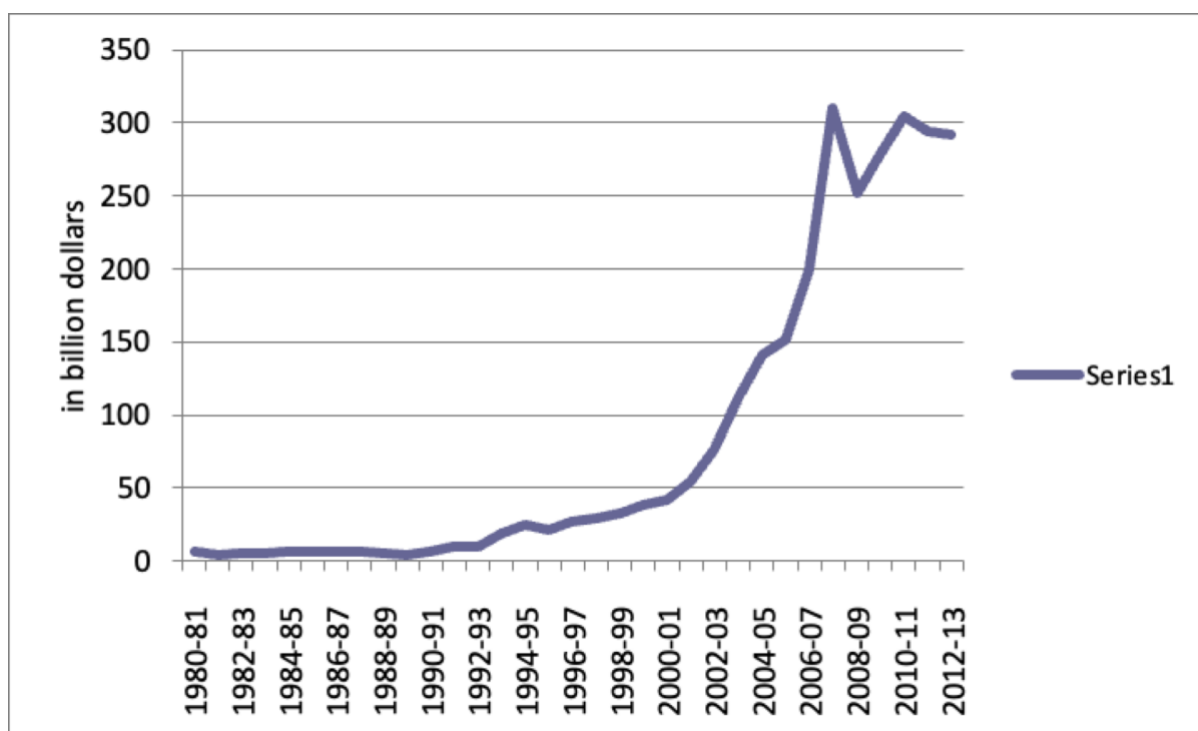
China's holdings of US Treasury Securities, in billions of US dollars

Source: <https://www.scmp.com/economy/china-economy/article/3213701/chinese-holdings-us-treasury-securities-falls-us8594-billion-lowest-2009-amid-rate-hikes-and-tension>



Dollar-Yuan Exchange Rate, 1981-2024

Source: <https://www.macrotrends.net/2575/us-dollar-yuan-exchange-rate-historical-chart> Dollar Yuan Exchange Rate - 35 Year Historical Chart



India's Forex Reserves in billion dollars

Source: Reserve Bank of India